

ORIE5270 - Big Data Technologies | ORIE6125 - Computational Methods in OR

Syllabus - Spring 2026

Last Updated: January 11th, 2026

Credit Hours: 2 credits (5270) | 3 credits (6125)

Class Meeting Time: MW 9:05-9:55am | MWF 9:05-9:55am

Location: Olin Hall 165 (MW), Olin Hall 165 (F)

Expected Supplemental Hours: Approximately 4 to 6 hours of supplementary work outside of the class per week (including practice problems, homework, and class project).

Instructor: Zhanhao Zhang

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Office hours: TBD.

Teaching Assistant: Qingyao Sun (qs234@cornell.edu), office hours TBD.

COURSE DESCRIPTION

This course offers a broad overview of computational techniques useful for data scientists. Topics include: unix shell, regular expressions, version control (git), data structures and algorithms, working with databases, data analysis using Python and related libraries (Pandas, NumPy/Scipy), parallel computing, data visualization, an overview of standard machine learning and optimization libraries in python (scikit-learn, pytorch).

Prerequisite(s): This course assumes that you have working knowledge of Python (fundamental Python semantics (e.g. shallow vs. deep copies), writing functions and classes, working with “core” libraries, and properly reading documentation).

COURSE COMMUNICATION

Communication for the course will be done primarily over e-mail and Ed. Major announcements (such as homework and grade releases) will be done over e-mail (via canvas). Ed Discussion is the place for students to ask technical and course logistic questions - other students are encouraged to answer questions they can (and will get credit towards their participation grade). With the exception of personal requests, all student questions should be posted on Ed.

HOMEWORK

Homeworks are a major component of your grade in both ORIE5270 and ORIE6125. They are normally due 2 or 3 weeks after their release and will be collected using Gradescope. If you fill out the course evaluation form at the end of the semester, **we will drop your lowest-graded homework when computing your final grade**. For each homework you will be able to submit regrade requests for **up to one week** after the grades are released.

Late submissions: You will get a total of **7 slip days** that you can use throughout the semester to submit late homeworks. If you submit late but within 24 hours of the deadline, you spend 1 slip day, if you submit within 48 hours you spend 2, and so on. **You are responsible for keeping track of your slip days** and making sure you have not gone over the 7 day limit. They are meant to help you in case of personal emergencies, travel, job interviews, and/or sickness, so avoid using them unless necessary.

FINAL PROJECT

A major component of the course is a final project where you (working in teams) will have the chance to combine tools and techniques we will go over throughout the semester. The goal is for you to create (or extend) a project utilizing several concepts, tools, and techniques covered in class. For example, the project could involve:

- Version control (preferably git)
- Unit testing for new features
- Documentation for any public APIs, if applicable
- HTML doc & visualization of the project

- Attention to performance, where relevant (you don't need to extensively tune your project but you should make e.g. informed decisions about the kind of data structures you use in core routines)

More details on dates and deliverables will be released on Canvas.

LEARNING OUTCOMES

The primary goal of this course is to introduce students to fundamental tools for working with data in python. The majority of the course will focus on *practical tools and best practices* for software development (think of it as a data science bootcamp!) that complement the theoretical tools taught in other ORIE masters level on machine learning and optimization. By the end of the course, students will be able to:

- Import data into a coding environment from a variety of sources including relational databases, flat files, APIs and from unstructured sources such as webpages.
- Write scalable code using best practices such as version control, unit tests, appropriate data structures, and parallel computing frameworks.
- Wrangle data in python using standard libraries and packages such as python, numpy, and scipy.
- Quickly visualize datasets using tools in python such as seaborn, plotly, and folium.
- Implement standard machine learning pipelines using scikit-learn, PyTorch.

GRADING POLICIES

The final grade for students in ORIE5270 will be calculated as follows:

Assignment	Grade Percentage
Homework	80%
Final Project	20%

The final grade for students in ORIE6125 will be calculated as follows and graded on a Sat/Unsat basis:

Assignment	Grade Percentage
Homework	50%
Seminar Presentation	10%
Final Project	40%

COURSE EXPECTATIONS

PERSONAL CONDUCT

The College of Engineering expects students to be respectful and professional in all participation and communication. You are expected to maintain professional conduct and speech in all aspects of this course. Professional behavior demands you have a responsible and mature attitude in person and online. Disrespectful, unethical, and/or unprofessional behaviors will not be tolerated and can result in course failure and/or dismissal from the program.

ACADEMIC STANDARDS

Each student is expected to abide by the Cornell University Code of Academic Integrity (<https://theuniversityfaculty.cornell.edu/dean/academic-integrity/code-of-academic-integrity/>). Work submitted by students for academic credit must be a student's own. Discussing problems with your peers at a high level is encouraged, *as long as you write your own solutions*. For example, you can discuss the high-level idea of an

implementation with a friend, but you should not copy or take notes of their (pseudo)code. If you have obtained your solution with the help of a classmate, a book or another source (i.e. ChatGPT, StackOverflow, a publication, technical blog/white paper) **you have to cite your source**. For written submissions this requires you to reference the source (using a citation style of your choice) at the point the ideas/solution were used. For code submissions this requires you to add a comment to your code at the point the idea/code begins being used with the source and another comment delineating when the referenced code ends. A sample reference snippet is included below:

```
##### Code for writing one-hot encoding taken from https://stackoverflow.com/questions/37292872/how-can-i-one-hot-encode-in-python
```

```
one_hot = pd.get_dummies(df['B'])
```

```
##### Referenced Code Ends
```

Violations of the Code of Academic Integrity, especially plagiarism, may result in a failing grade in the course. Students are urged to read and complete the exercises on “Recognizing and Avoiding Plagiarism” at: <https://plagiarism.arts.cornell.edu/tutorial/index.cfm>

INCLUSIVE LEARNING ENVIRONMENT

Cornell supports an inclusive learning environment where diversity and individual differences are understood, respected, appreciated, and recognized as a source of strength. It is expected that students in this class will respect differences and demonstrate diligence in understanding how other peoples' perspectives, behaviors, and worldviews may be different from their own.

ACCOMMODATIONS FOR STUDENTS WITH DISABILITIES

Cornell University is committed to full inclusion for all persons to its educational program and services. Services and reasonable accommodations are available to persons with temporary and permanent disabilities when conditions cause barriers to equal educational opportunity. Student Disability Services (SDS) determines the eligibility of students to receive accommodations and works collaboratively with the student and university faculty and staff to recommend appropriate accommodations. Students are advised to contact SDS as early as possible in the semester to ensure appropriate accommodations: www.sds.cornell.edu, 607-254-4545.

ATTESTATION

By registering for this class and accessing course materials through Canvas, students agree to abide by University, College, Department, and Course policies.